



Aror University of Art, Architecture, Design & Heritage Sukkur.

Department of Artificial Intelligence and Multimedia Gaming

Programming for AI (Spring-2026)

LAB No. 08

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Objective of Lab No. 08:

After performing lab 8, students will be able to:

1. Establish Database Connectivity with Python
2. Perform DDL and DML Operations using Python

Lab Tasks:

Task 1: The "Eco-Track" Inventory System

You have been hired by a local sustainable grocery store to manage their inventory. The owner wants to track products, their categories, and stock levels.

1. Database Setup: Create a database named `EcoTrackDB` and a table `Products` (id, name, price, stock_quantity) and a table `Categories` (cat_id, cat_name).
2. Insert five products and two categories.
3. Write a query to display all products along with their category names using an `INNER JOIN`.
4. The store just received a shipment; update the `stock\_quantity` for "Apples" by adding 50 units.
5. The store will no longer carry "Plastic Straws." Remove this record from the database.



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7. Display those products which are expensive (Price above 500)

Task 2: "Swift-Stay" Hotel Management:

A boutique hotel needs a system to manage guest bookings. You need to handle the relationship between guests and the rooms they reserve.

1. Create `SwiftStayDB`. Create a `Guests` table (guest_id, name, email) and a `Bookings` table (booking_id, guest_id, room_number, check_in_date).
2. Insert three guests and four booking records.
3. Retrieve only the names and emails of guests who have a booking for "Room 101."
4. Perform a `LEFT JOIN` to show all guests, even those who don't have a current booking.
5. A guest changed their email address; update the `email` field for a specific `guest\_id`.
6. A booking was canceled; delete the specific booking record without removing the guest from the system.

Task 3: "Edu-Link" University Portal:

A university department wants a tool to link students to the specific courses they are enrolled in.

1. Create `EduLinkDB`. Create a `Students` table (student_id, name, major) and a `Courses` table (course_id, course_title, instructor).
2. Insert data for six students across different majors and three courses.
3. Select/Filter: Select all students who are majoring in "Computer Science."



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4. Use a `JOIN` to create a "Class Roster" that lists student names alongside the titles of the courses they are taking.
5. Use a command to `DROP` a temporary table you created for "Testing" purposes earlier in the script.
6. The instructor for "Database 101" has changed; update the `instructor` name in the `Courses` table.